模拟真实世界：多模态生成式模型的统一综述

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摘要—在通用人工智能（AGI）研究中，理解和复现真实世界是一项关键挑战。为了实现这一目标，许多现有方法（如世界模型）旨在捕捉支配物理世界的根本原理，从而实现更精确的仿真和有意义的交互。然而，当前的方法通常将不同模态（包括 2D 图像、视频、3D 以及 4D 表示）视为独立的领域，忽视了它们之间的相互依赖关系。此外，这些方法往往只关注现实的孤立维度，而未能系统性地整合其内在联系。在本综述中，我们提出了一个针对多模态生成式模型的统一视角，探讨真实世界仿真中数据维度演进的过程。具体而言，本综述从 2D 生成（外观）开始，继而过渡到视频生成（外观+动态）与 3D 生成（外观+几何），最终迈向集成所有维度的 4D 生成。据我们所知，这是首次尝试在一个统一框架内系统性地整合 2D、视频、3D 和 4D 生成的研究。为引导未来研究，我们全面回顾了相关数据集、评估指标及未来方向，以帮助新手获得深入洞见。本综述旨在搭建一座桥梁，推动多模态生成式模型与真实世界仿真在统一框架下的发展。

***Index Terms*—生成式模型、图像生成、视频生成、3D 生成、4D 生成、深度学习、文献综述。**

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A project associated with this survey is available at https://github.com/ALEEEHU/World-Simulator.

I. Introduction

F数十年来，研究界一直致力于开发能够体现物理世界基本原理的系统，这是迈向通用人工智能（AGI）旅程中的基石 [1]。这一努力的核心在于利用机器模拟真实世界，旨在通过多模态生成式模型的视角捕捉现实的复杂性。由此产生的世界模拟器有望推动对真实世界的理解，解锁虚拟现实 [2]、游戏 [3]、机器人 [4] 以及自动驾驶 [5] 等变革性应用。

术语“世界仿真器”最早由 Ha David [6] 提出，将其类比于认知科学中的“心智模型” [7] 概念。在此基础上，现代研究者 [8] 将仿真器形式化为一种抽象框架，使智能系统能够通过多模态生成式模型来仿真真实世界。这些模型将真实世界的视觉内容和时空动态编码为紧凑的表示。由于几何、外观和动态共同决定了生成内容的真实感，这三个方面被社区广泛研究 [9]。传统的现实世界仿真方法长期依赖于结合几何、纹理和动态的图形技术。具体而言，几何和纹理建模 [10] 用于创建物体，而关键帧动画 [11] 和基于物理的仿真 [12] 等方法则用于模拟物体随时间的运动与行为。

尽管取得了巨大进展，这些传统方法通常需要大量手动设计、启发式规则定义以及计算成本高昂的处理过程，限制了其在多样化场景中的可扩展性和适应性。近年来，基于学习的方法，特别是多模态生成式模型，通过提供数据驱动的现实仿真方式，彻底改变了内容创作。这些方法减少了对人工干预的依赖，提升了任务间的泛化能力，并实现了人与模型之间的直观交互。例如，Sora [13] 因其出色的仿真能力而受到广泛关注，展示了对物理规律的初步理解。这类生成式模型的出现带来了新的视角和方法论，通过减少对复杂手动设计和高计算

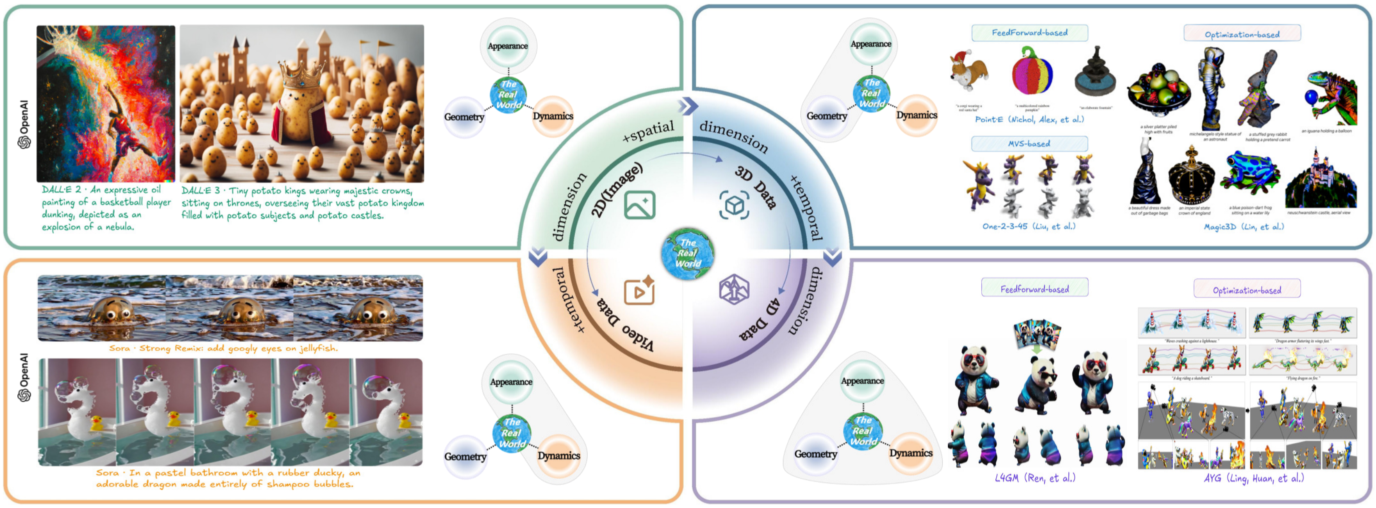

图 1: 从二维图像到视频、三维及四维内容在真实世界仿真中的维度增长路线图，基于数据属性（即外观、几何和动态）的覆盖范围，勾勒出一个概念分类体系。

成本建模的需求，同时增强了在多样化仿真场景中的适应性和可扩展性，从而克服了传统方法的局限性。

尽管现有的生成式模型在不同数据维度上提供了强大的真实内容合成技术，但现实世界本身具有固有的高维复杂性，目前仍缺乏对这些跨维度进展进行系统整合的全面综述。本综述旨在通过数据维度增长的视角，弥合这一差距，如图 1 所示。具体而言，我们从二维生成（仅外观）出发，随后通过引入动态性和几何性维度，分别扩展至视频生成和三维生成。最后，通过整合所有维度，实现四维生成。

总而言之，本综述做出了三项关键贡献。首先，它从数据维度增长的角度，通过多模态生成式模型的视角，对现实世界仿真的方法进行了系统性回顾。据我们所知，这是首个将 2D、视频、3D 和 4D 生成统一研究的综述，为该研究领域提供了结构化且全面的概述。其次，它从多个角度调研了常用的数据集、其特性以及相应的评估指标。第三，它指出了开放的研究挑战，旨在引导该领域的进一步探索。

以往对生成模型的综述主要分别关注文本到图像、文本到视频以及文本到 3D 生成，而未全面研究它们之间的关系。与此不同，本综述旨在通过追踪生成模型如何从仅处理外观（2D 生成）逐步发展到融入动态性（视频生成）、几何结构（3D 生成），最终实现外观、动态性和几何结构在 4D 生成中的融合，为多模态生成模型提供更综合的视角。这种维度视角旨在弥合此前孤立的研究领域，并揭示它们之间共有的挑战与机遇。

我们期望本综述能够为新手提供有价值的见解，并促进资深研究者进行批判性分析。本文综述的结构如下：第 II 节介绍深度生成模型的基础概念。第 III 节阐述四个关键范式：2D、视频、3D 以及 4D 生成。第 IV 节回顾这些范式的数据集与评估指标。最后，第 V 节概述未来发展方向，第 VI 节对综述进行总结。

II. 初步研究

深度生成模型借助深度神经网络学习复杂且高维的数据分布。将数据样本记为 $\mathbf{x}$，其分布记为 $p_{data}(\mathbf{x})$，深度生成模型的目标是用 $p_\theta(\mathbf{x})$ 近似 $p_{data}(\mathbf{x})$，其中 θ 为模型的参数。在本节中，我们简要回顾几种主流的生成模型（表 I），包括生成对抗网络（GANs） [14]、变分自编码器（VAEs） [15]、自回归模型（AR Models） [16]、归一化流（NFs） [17] 以及扩散模型 [18]。

A. 生成对抗网络（GANs）

GANs 避免了 $p_\theta(\mathbf{x})$ 的参数形式，而是将 $p_\theta(\mathbf{x})$ 表示为生成器产生的样本的分布。在某些条件下，已证明 $p_\theta(\mathbf{x})$ 将收敛到 $p_{data}(\mathbf{x})$ [14]。

具体而言，生成器将一个噪声向量 $\mathbf{z}$ 作为输入，以合成一个数据样本 $G(\mathbf{z};\theta_g)$。其中，$p_\theta(\mathbf{x})$ 定义为 $G(\mathbf{z};\theta_g)$ 的分布，即 $\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})$。同时，判别器 $D(\mathbf{x};\theta_d)$ 用于判断输入的数据样本是真实的还是合成的。在训练过程中，判别器被训练以区分生成的样本与真实数据，而生成器则被训练以欺骗判别器。该过程可表述为：

$$\min_G \max_D V(G,D) = \mathbb{E}_{\mathbf{x}\sim p_{data}(\mathbf{x})}[\log D(\mathbf{x};\theta_d)] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1-D(G(\mathbf{z};\theta_g);\theta_d))]. \tag{1}$$

表 I: 深度生成模型的比较。

Model Types	Advantages	Disadvantages
GANs [14]	1) Flexible 2) Efficient inference	1) Unstable training 2) Lack diversity 3) Mode collapse
VAEs [15]	1) Data compression	1) Posterior collapse 2) Blur
AR Models [16]	1) Explicit density	1) Markov assumption 2) Difficult to parallelize
NFs [17]	1) Explicit density	1) Limited capacity 2) Restricted architecture
Diffusion Models [18]	1) High-quality samples 2) Learn complex distribution	1) Expensive computation

生成对抗网络训练存在多个挑战。例如，纳什均衡可能并不存在 [19] 或难以达到 [20]，导致训练不稳定。另一个问题是模式崩溃，即生成器仅产生特定类型的样本，多样性较低 [20], [21]。

B. 变分自编码器（VAEs）

变分自编码器（VAEs）将 $p_\theta(\mathbf{x})$ 表述为，

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})d\mathbf{z}, \tag{2}$$

其中 $p_\theta(\mathbf{z})$ 是 $\mathbf{z}$ 的先验分布，$p_\theta(\mathbf{x}|\mathbf{z})$ 是在 $\mathbf{z}$ 条件下 $\mathbf{x}$ 的分布。然而，由于该积分通常难以处理，变分自编码器（VAEs）最大化 $\log p_\theta(\mathbf{x}) \geq -\mathtt{KL}(q_\theta(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) + \mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]$ 的下界，其中 $\mathtt{KL}(q_\theta(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$ 为 $q_\theta(\mathbf{z}|\mathbf{x})$ 与 $p_\theta(\mathbf{z})$ 之间的 KL 散度，$\mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]$ 由随机梯度变分贝叶斯估计量 [15] 计算得出。变分自编码器的挑战包括不理想的稳定平衡、模糊性等问题 [22]。

C. 自回归模型（AR 模型）

自回归（AR）模型将 $p_\theta(\mathbf{x})$ 分解为条件概率的乘积，

$$p_\theta(\mathbf{x}) = p(x_1, \dots, x_d) = \prod_{i=1}^{d} p_\theta(x_i \mid x_1, \dots, x_{i-1}), \tag{3}$$

其中 d 为序列长度。这种因子分解简化了多元密度估计，并已被广泛用于以顺序方式建模图像中的像素 [23]–[25]。

为了降低标准基于 Transformer 的自回归模型中注意力机制的二次计算开销，近期已引入了几种非 Transformer 架构。RWKV [26]、Mamba [27] 和 RetNet [28] 用循环或状态空间机制替代或增强注意力机制。RWKV 和 Mamba 采用纯粹的循环设计，维持固定大小的内存，在中等序列长度下实现线性时间推理，但在极端上下文规模下仍面临挑战。RetNet 通过保留机制更新隐状态，为全局自注意力提供了一种高效替代方案。尽管这些架构在语言及其他序列任务上表现出色，但其作为深度生成模型主干网络的应用仍有限。未来将它们整合到生成流水线中的工作，有望改善样本质量、可扩展性与内存使用之间的权衡。

D. 归一化流（NFs）

NFs 采用一个可逆神经网络 $g(\cdot)$ 将 $\mathbf{z}$ 从已知且易处理的分布映射到真实数据分布。通过这种方式，$p_\theta(\mathbf{x})$ 可以表示为，

$$p_\theta(\mathbf{x}) = p(f(\mathbf{x}))|\mathtt{det\ J}(g(g^{-1}(\mathbf{x})))|^{-1}, \tag{4}$$

其中 $g^{-1}(\cdot)$ 是 $g(\cdot)$ 的求逆，$\mathtt{J}(g(\mathbf{z})) = \frac{\partial g}{\partial \mathbf{z}}$ 是 $\mathbf{g}$ 的雅克比。NFs 通过组合一组 N 双射函数构造更复杂的非线性可逆函数，并定义 $g = g_N \circ \cdots \circ g_1$。

E. 扩散模型

扩散模型是一类概率生成模型，通过逐步引入噪声来破坏数据，并随后学习逆转这一过程以生成样本。我们定义 $p_\theta(\mathbf{x})$ 时包含一个能量项 $s_\theta(\mathbf{x})$ ，

$$p_\theta(\mathbf{x}) = \exp(-s_\theta(\mathbf{x}))/Z_\theta, \tag{5}$$

其中 $Z_\theta = \int \exp(-s_\theta(\mathbf{x}))d\mathbf{x}$ 是规范化项。由于 Z_θ 难以计算，扩散模型改为学习评分函数 $\nabla_\mathbf{x} \log p_\theta(\mathbf{x}) = -\nabla_\mathbf{x} s_\theta(\mathbf{x})$ 。

将数据分布变换为标准高斯分布的前向过程由 $d\mathbf{x} = f(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ 定义，其中 $f(\mathbf{x}, t)$ 为漂移系数，$g(t)$ 为扩散系数，$\mathbf{w}$ 为标准维纳过程，$t \in [0, 1]$。样本通过对应的逆向过程生成，该过程由 $d\mathbf{x} = [f(\mathbf{x}, t) - g^2(t)\nabla \log p_t(\mathbf{x})]dt + g(t)d\overline{\mathbf{w}}$ 描述，其中 $\overline{\mathbf{w}}$ 为时间从 1 到 0 反向流动时的标准维纳过程。

III. 范式

本节从数据维度增长的角度，介绍了模拟现实世界的方法。首先介绍二维生成（第 III-A 节）用于外观建模，然后通过引入动态和几何维度，依次拓展到视频生成（第 III-B 节）和三维生成（第 III-C 节）。最后，通过整合这三维信息，展示了近期在四维生成（第 III-D 节）方面的进展。

A. 二维生成

近年来，生成式模型领域取得了显著进展，尤其是在文本到图像生成方面。文本到图像生成因其能够根据文本描述生成逼真的图像而受到关注，这些图像能够捕捉现实世界的外观特征。通过利用扩散模型、大模型（LLMs）和自编码器等技术，这些模型实现了高质量且语义准确的图像生成。

1) 算法: **Imagen** [29] 基于 GLIDE 所确立的原则，但引入了显著的最优化和改进。与从头开始训练特定任务的文本编码器不同，Imagen 采用预训练且冻结的语言模型，从而降低了计算需求。Imagen 测试了在图像-文本数据集（例如，CLIP [30]）上训练的模型，以及在纯文本数据集（例如，BERT [31] 和 T5 [32]）上训练的模型。这一做法表明，通过缩放语言模型能够比扩大图像扩散模型更有效地提升图像保真度和文本一致性。

DALL-E [33]（版本 1）采用 Transformer 架构，将文本和图像作为单一数据流进行处理。DALL-E 2 [34] 利用 CLIP [30] 强大的语义和风格能力，其采用生成式扩散解码器来逆转 CLIP 图像编码器的流程。DALL-E 3 [35] 在 DALL-E 2 [34] 的基础上进一步发展，显著提升了图像保真度和文本对齐能力。它增强了文本理解，使得从复杂描述中生成更准确、更细腻的图像成为可能。DALL-E 3 与 ChatGPT [36] 集成，使用户能够在 ChatGPT 界面中直接构思和优化提示词，简化了生成详细且定制化提示词的过程。该模型生成的图像具有更高的真实感，并与所提供文本更好地对齐，使其成为创意和专业应用的强大工具。

DeepFloyd IF [37] 以其卓越的逼真度和先进的语言理解能力而著称。该系统采用模块化设计，包含一个静态文本编码器和三个顺序排列的像素扩散模块。初始阶段，基础模型根据文本描述生成 64× 64 像素的图像。随后，通过两个超分辨率模型将图像分别提升至 256× 256 像素和 1024× 1024 像素。每个阶段均使用源自 T5 [32] Transformer 的静态文本编码器生成文本嵌入，这些嵌入随后由带有集成交叉注意力和注意力池化机制的 U-Net 架构进行处理。

Stable Diffusion(SD) [38]，也称为潜在扩散模型（Latent Diffusion Model, LDM），在计算资源有限的情况下仍能提升训练与推理效率，并生成高质量且多样化的图像。降噪过程发生在预训练自编码器的潜在空间中，该自编码器将图像映射到一个空间潜在表示。其潜在的 U-Net 架构通过引入交叉注意力机制来建模条件分布，该条件可包含文本提示、分割掩码等。它使用 CLIP [30] 文本嵌入作为条件，并在 LAION [39] 数据集上进行训练，以生成分辨率为 512× 512 的图像（潜在分辨率为 64× 64）。基于 Stable Diffusion，SDXL [40] 采用了三倍于原模型大小的 U-Net 骨干网络，引入了额外的注意力块，并通过使用第二个文本编码器扩大了交叉注意力上下文。此外，SDXL 还包含一个精炼模型，通过后处理的图像到图像技术提升由 SDXL 生成样本的视觉保真度。

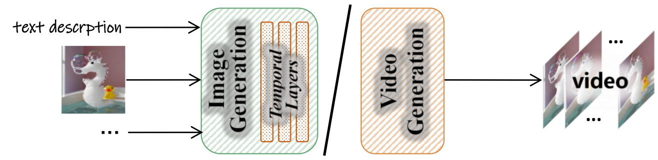

图 2: 视频生成范式的示意图。视频生成模型在图像生成模型的基础上，通过添加时间层或从零开始构建而成。

FLUX.1 [41] 采用了一种混合架构，该架构整合了多模态与并行扩散 Transformer 块，实现了高达 120 亿参数的惊人规模。通过采用流匹配（flow matching）这一简单而有效的生成式模型训练技术，FLUX.1 在性能上超越了先前的最先进扩散模型。该套件还包含旋转位置嵌入和并行注意力层，显著提升了模型性能与效率。

B. 视频生成

由于图像与视频在结构上的相似性，早期的视频生成方法主要采用现有 2D 图像生成模型的适配和微调（第 III-A 节）。最初的研究视角集中在引入时序动态建模机制，通常通过在架构中添加时序层（图 2），如注意力和卷积层，来实现。一种常见的训练策略还涉及混合使用 2D 图像和视频数据，以提升生成视频的视觉质量。

受 Sora [13] 的启发，当前最先进的模型普遍采用扩散 Transformer 架构。这些模型在压缩的时空潜在空间中运行，将视频分解为一系列“补丁”，作为 Transformer 的 token。该方法能够同时处理空间和时序信息。尽管这些模型具有高度复杂性，但它们通常仍保留处理单张图像（即单帧视频）的能力，从而可以利用大量现有的 2D 图像生成数据。

在本节中，我们根据其潜在的生成式机器学习架构，将这些模型分为三大类别。图 3 总结了近期的文本到视频生成技术。对于希望进行更深入探索的读者，相关详细综述可参考 [42], [43]。

1) 算法: **(1) 基于变分自编码器和生成对抗网络的方法**。在扩散模型出现之前，视频生成研究主要通过两种方法取得进展：基于变分自编码器（VAE）和生成对抗网络（GAN）的方法，它们各自为视频合成的挑战提供了独特的解决方案。基于 VAE 的方法从随机动力学的 SV2P [44] 发展到将 VQ-VAE [71] 与 Transformer 结合的 VideoGPT [72]，通过分层离散潜变量高效处理高分辨率视频。显著改进体现在 FitVid [45] 中参数高效的架构设计，以及引入对抗训练以实现更真实的预测。与此同时，基于 GAN 的方法也取得了重要进展，始于 MoCoGAN [46]，该方法将内容与运动成分分解以实现可控生成。StyleGAN-V [47] 进一步将其扩展，通过位置嵌入将视频视为时间连续信号，而 DIGAN [48] 则引入隐式神经表示以提升连续视频建模效果。StyleInV [49] 利用预训练的 StyleGAN [73] 生成器，结合时序风格调制的反演网络，在保持时间连贯性的前提下实现了高质量帧合成，标志着又一里程碑。

(2) Diffusion-based Approaches. 文本到视频生成近年来取得了显著进展，相关方法大致可分为两类：基于 U-Net 的架构和基于 Transformer 的架构。

(i) 基于 U-Net 的架构。开创性的视频扩散模型（VDM）[50] 通过扩展图像扩散架构并引入联合图像-视频训练，实现了高保真、时间连贯的视频生成，同时降低了梯度方差。Make-A-Video [51] 在无需成对文本-视频数据的情况下，通过利用现有的视觉表示 [30] 和创新的时空模块，推进了文本到视频的生成。Imagen Video [52] 引入了级联扩散模型，结合基础生成与超分辨率，而 MagicVideo [53] 则通过在低维潜在空间中使用潜在扩散实现了高效生成。GEN-1 [54] 专注于基于深度估计的结构保持编辑，PYoCo [55] 通过精心设计的视频噪声先验，在数据有限的情况下展示了高效的微调能力。Align-your-Latents [56] 通过扩展 Stable Diffusion [38] 的时间对齐技术，实现了高分辨率生成（1280×2048）。Show-1 [74] 结合了基于像素和基于潜在的方法，以提升质量并减少计算量。VideoComposer [57] 提出了一种新颖的可控合成范式，通过时空条件编码器实现灵活组合，可基于多种条件进行创作。AnimateDiff [58] 提出了即插即用的运动模块，具备可迁移的运动先验，并引入 MotionLoRA 实现高效适应。PixelDance [59] 通过结合首帧和末帧图像指令以及文本提示，提升了生成效果。

(ii) 基于 Transformer 的架构。在扩散 Transformer（DiT）[75] 取得成功之后，基于 Transformer 的模型逐渐受到关注。VDT [62] 为包括预测、插值和补全在内的多种任务引入了模块化的时空注意力机制。W.A.L.T [63] 通过统一的潜在空间和因果编码器架构实现了逼真的图像生成，能够在 512×896 生成高分辨率视频。Snap Video [76] 通过处理空间和时间上的冗余像素，将训练效率提升了 3.31 倍；而 GenTron [64] 在无需运动引导的情况下扩展至超过 30 亿参数。Luminia-T2X [65] 通过零初始化注意力和分词化的潜在时空空间，整合了多种模态信息。CogVideoX [66] 通过专家 Transformer、三维变分自编码器（3D VAEs）以及渐进式训练，在长时视频生成方面表现出色，其性能经多个指标验证达到了当前最优水平。

突破性的 Sora [13] 是一种先进的扩散 Transformer 模型，强调在不同分辨率、宽高比和持续时间下生成高质量图像与视频。Sora 通过将潜在时空空间进行分词化，实现了灵活且可扩展的生成能力。

(3) Autoregressive-based Approaches. 与基于扩散的方法并行，受大模型（LLMs）启发的自回归框架已成为视频生成的一种替代方法。这些方法通常遵循两阶段流程：首先使用向量量化自编码器（如 VQ-GAN [77] 和 MAGVIT [68], [78]–[81]）将视觉内容编码为离散的潜在 token，然后对潜在空间中的 token 分布进行建模。

CogVideo [69] 是一个基于预训练文本到图像模型 CogView [82] 构建的 90 亿参数 Transformer 模型，在这一方向上取得了重要进展。它采用多帧率分层训练策略以增强文本与视频的对齐能力，并作为首批开源的大规模预训练文本到视频模型之一，在机器和人类评估中均建立了新的基准。

VideoPoet [70] 提出了一种仅解码器的 Transformer 架构，用于 zero-shot 视频生成，能够处理包括图像、视频、文本和音频在内的多种输入模态。遵循大模型训练范式，包含预训练和任务特定适应两个阶段，VideoPoet 在 zero-shot 视频生成方面达到了当前最优性能，尤其在运动保真度方面表现突出，得益于其多样化的生成预训练目标。

评估。视频生成模型的评估随着任务复杂性的增加而不断发展。早期的方法依赖于基于分布的度量，其中

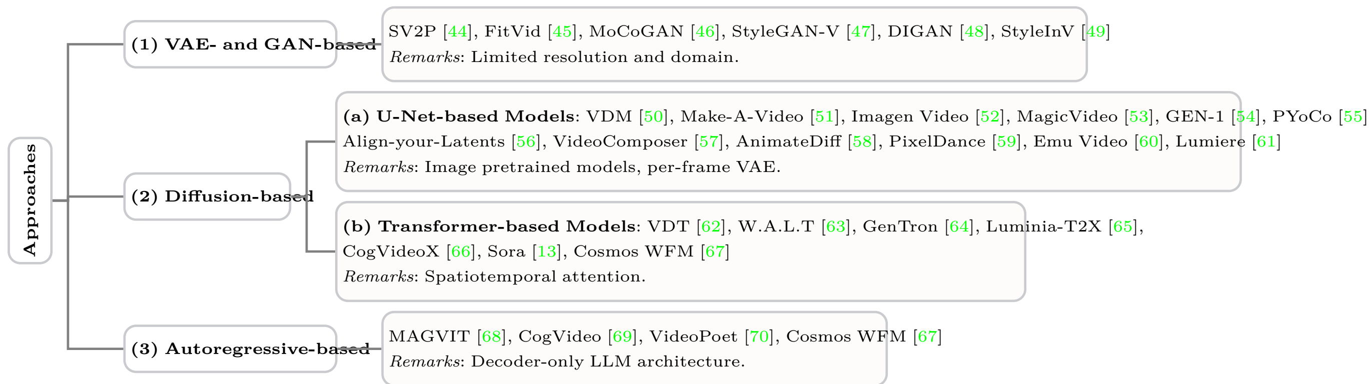

图 3: 文本到视频生成技术的概述，按三种主要方法进行分类。

Dimension (↑)	Hunyuan [87]	Mochi [88]	SoRA [13]	Wan-14B [89]
Motion Quality	0.413	0.420	0.482	0.415
Human Artifacts	0.734	0.622	0.786	0.691
Pixel Stability	0.983	0.981	0.952	0.972
ID Consistency	0.935	0.930	0.925	0.946
Physical	0.898	0.728	0.933	0.939
Smoothness	0.890	0.530	0.930	0.910
Image Quality	0.605	0.530	0.665	0.640
Scene Quality	0.373	0.368	0.388	0.386
Stylization	0.386	0.403	0.606	0.328
Object Accuracy	0.912	0.949	0.932	0.952

表 II: 不同文本到视频方法的定量比较。结果来自 [89]。

最著名的是弗雷歇视频距离（Fréchet Video Distance, FVD） [50], [56], [83]。作为图像领域弗雷歇初始距离（FID） [84] 的时序扩展，FVD 通过比较时空特征分布来评估视觉质量和连贯性。近年来，诸如 VBench [85] 等基准测试提供了对运动平滑性、主体身份等特定属性的更细致分析，利用 CLIP [30] 和 DINO [86] 等模型的特征。然而，由于自动化度量常常与人类感知不一致，该领域越来越多地转向人类研究，以实现更全面和准确的评估，尤其是在先进开放领域模型方面。表 II 展示了现代视频生成模型的人类偏好评估结果。

2) 应用: **(1) Video editing** 近期在扩散模型的推动下取得了显著进展，能够在保持时间一致性的同时实现复杂的修改。该领域通过多种创新方法不断发展，针对视频操作的不同方面提出了有效解决方案。

早期的工作包括 Tune-A-Video [90], 其开创了 one-shot 调整范式，通过时空注意力机制将文本到图像的扩散模型扩展至视频生成。

时间一致性问题已通过多种方法得到解决。Vid-ToMe [91] 引入了 token 合并以对齐帧，而 EI [92] 则开发了专用的注意力模块。

若干研究聚焦于特定编辑能力。Ground-A-Video [93] 通过基于定位引导的框架解决了多属性编辑问题，而 Video-P2P [94] 则引入了交叉注意力控制以实现角色生成。

近期的框架如 UniEdit [95] 和 AnyV2V [96] 代表了最新演化方向，提供了无需调参的方法和简化的编辑流程。专门的应用如 CoDeF [97] 和 Pix2Video [98] 引入了创新技术，用于实现时间一致性的处理以及渐进式变化传播。

这些方法成功在内容编辑与结构保留之间取得平衡，标志着视频操作技术的重大进步。

(2) Novel view synthesis 已被视频扩散模型彻底革新，这些模型得益于对真实世界几何结构的学成先验，能够从有限的输入图像中生成高质量视角。ViewCrafter [99] 首次开创了这一方向，将视频扩散模型与基于点的三维表示相结合，引入了迭代合成策略和相机轨迹规划，从而实现从稀疏输入中生成高保真结果。相机控制已成为关键方面，CameraCtrl [100] 通过即插即用模块实现了精确的相机位姿控制。若干创新方法解决了视角一致性挑战。ViVid-1-to-3 [101] 将新视角生成重新定义为相机运动的视频生成，而 NVS-Solver [102] 引入了一种 zero-shot 范式，通过给定视角调制扩散采样过程。这一趋势表明，正朝着利用视频扩散先验的同时保持几何一致性和相机控制的方向收敛，从而推动日益逼真的合成应用发展。

(3) Human animation in videos 在视频生成中获得了重要地位，这在世界仿真器中扮演着关键角色，如第 III-B1 节所述。这一点尤其重要，因为人类是现实世界中最核心的参与者，使其真实模拟至关重要。

得益于生成式模型早期的成功，一些代表性工作 [46], [103], [104] 引入了生成对抗网络（GAN） [14]

来实现视频中人物的动画化。尽管取得了这些进展，人物视频动画中最核心的问题仍然是生成视频的视觉保真度。ControlNet [105] 和 HumanSD [106] 是基于基础文本到图像模型（如 Stable Diffusion [38]）参考姿态进行人物动画的即插即用方法。

此外，为解决这些方法的泛化问题，animate-anyone [107] 提出了一种 ReferenceNet，以保持参考视频更多的空间细节，并将野外生成质量推至新里程碑。一些后续工作 [108], [109] 基于 animate-anyone 尝试简化训练架构和成本。

另外，随着计算机图形学领域对几何与纹理的深入研究，一些工作将 3D 建模引入人物视频动画中。Liquid Warping GAN [110]、CustomHuman [111] 以及 LatentMan [112] 是早期尝试将 3D 人体先验引入生成环路的代表。最新进展 MIMO [113] 则分别显式建模角色、3D 运动和场景，以驱动野外环境中的人物动画。这些方法，无论是否包含 3D 先验，都极大地推动了将人类引入世界仿真器循环的重要一步。

C. 3D 生成

3D 生成关注几何与外观两个方面，以更好地模拟真实世界场景。在本节中，我们探讨了各种 3D 表示方法和生成算法，对近期进展进行了系统性综述。具体而言，我们根据输入模态将 3D 生成方法进行分类，包括*文本到 3D 生成*，即直接从文本描述合成 3D 内容；*图像到 3D 生成*，通过引入图像约束来优化基于文本驱动的输出；以及*视频到 3D 生成*，利用视频先验实现更一致的 3D 生成。图 7展示了这些进展的时间线总结，而表 IV则提供了前沿方法的全面对比。值得注意的是，部分方法跨越多个类别，体现了现代 3D 生成技术的多样性。

与从头构建 3D 生成模型不同，大多数现有方法高度耦合于 2D 和视频生成模型，以利用其强大的外观建模能力来提升 3D 生成效果，如图 5、8和10所示。首先，2D 生成模型中编码的图像先验以及视频生成模型中编码的几何线索可用于为 3D 生成模型提供监督信号。其次，可以对 2D 和视频生成模型进行微调，使其额外接收 3D 信息（例如法向量）作为输入，以合成具备 3D 感知能力的多视角图像，从而促进 3D 生成。

1) 3D 表示: 在 3D 生成领域，选择最优的 3D 表示方法至关重要。对于神经场景表示，3D 数据通常可以分为三类主要表示方法：显式、隐式和混合表示，如图 4所示。

(1) 显式表示。显式表示能够提供对物体和场景的精确可视化，由一组元素定义。传统的形式，如点云、网格和体素，多年来一直被广泛使用。

(i) 点云是无序的 3D 点集 (x, y, z)，通常还包含颜色或法向量等属性。由于深度传感器输出的广泛使用，点云被广泛应用，但其不规则结构 [120], [121] 对传统二维领域的神经网络提出了挑战。

(ii) 体素网格由称为体素的基本元素组成。体素作为像素 [122] 的三维对应物，代表三维网格上的点。它将一个包围盒划分为更小的元素，每个元素具有一个占据值。虽然体素可以存储诸如不透明度或颜色之类的数据，但由于立方体内存增长（$\mathcal{O}(n^3)$）和稀疏占据，它们在高分辨率数据下内存效率低下。

(iii) 网格由顶点和边构成多边形（面），定义三维空间中的二维表面。它们比体素网格更节省内存，仅编码物体表面；与点云不同，网格提供了显式的连接性，支持几何变换和高效的纹理编码。

(iv) 3D 高斯点阵（3DGS）被提出作为一种有效的方法，用于加速训练和渲染任务 [116]。该技术将物体建模为各向异性高斯分布的集合，其中每个分布由其位置（即均值 $\mathbf{x} \in \mathbb{R}^3$）、协方差矩阵 、不透明度 $\alpha \in \mathbb{R}$ 以及球谐函数系数 $\mathcal{C} \in \mathbb{R}^k$（其中 k 表示自由度）定义，从而能够建模视角相关的颜色变化，其表达式为：

$$G(X) = e^{-\frac{1}{2}\mathbf{x}^T\ ^{-1}\mathbf{x}}. \tag{6}$$

为了便于最优化，协方差矩阵 通常被分解为一个缩放矩阵 $\mathbf{S}$ 和一个旋转矩阵 $\mathbf{R}$ ，使得：

$$\Sigma = \mathbf{R}\mathbf{S}\mathbf{S}^T\mathbf{R}^T. \tag{7}$$

(2) 隐式表示。隐式表示使用连续函数（如数学模型或神经网络）描述三维空间，捕捉体素属性而非表面几何结构。隐式神经表示利用神经网络近似这些函数，以更高的训练和推理开销换取更强的表达能力。主要方法包括带符号距离场（SDF）[117] 和神经辐射场（NeRF）[123]。

(i) 带符号距离场 (SDF) 通过神经场表示三维形状，其中表面被隐式定义为 SDF 的零值等高线。给定一个点 $\mathbf{x} \in \mathbb{R}^3$ ，SDF 函数 $f(\mathbf{x}) = d$ 返回该点到最近表面点的最短距离，符号表示 $\mathbf{x}$ 位于形状内部（$d < 0$ ）还是外部（$d > 0$ ）。表面由满足 $f(\mathbf{x}) = 0$ 的点集定义。SDF 的准确率依赖于精确的法向量，因为其方向决定了距离的符号，因此准确估计法向量对于表面表示至关重要。

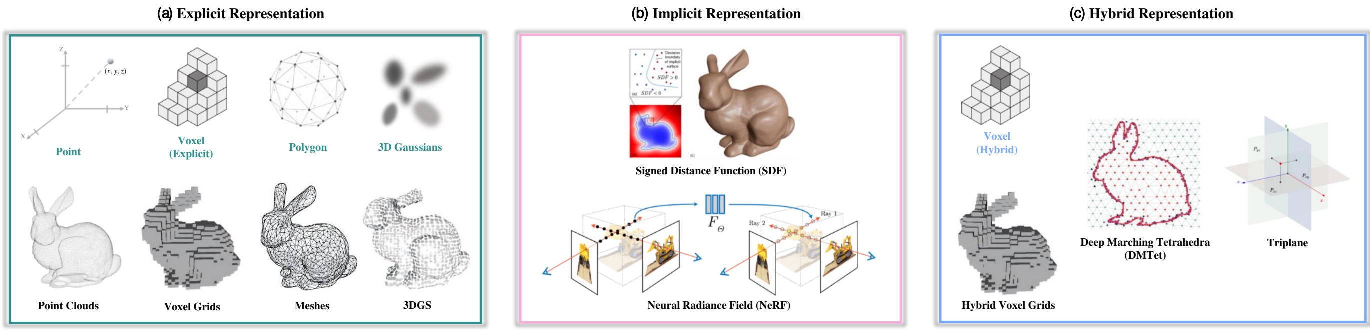

图 4: 神经场景表示的三大类别。(a) 显式表示通过点云、体素网格 [114]、网格 [115] 和 3D 高斯 [116] 直接存储几何信息。(b) 隐式表示通过函数（如带符号距离函数 SDF [117] 和神经辐射场 NeRF [118]）定义物体，能够在不固定分辨率的情况下实现平滑连续的表面。(c) 混合表示结合显式与隐式方法，采用混合体素网格、深度行走四面体 (DMTet) [119] 和三平面等技术，以获得更高的效率和灵活性。

(ii) 神经辐射场（NeRF） [118], [123] 将 3D 场景表示为编码在多层感知机（MLP）权重中的连续体素函数。与 DeepSDF [117] 不同，NeRF 回归的是密度和颜色，而非带符号距离函数，从而实现体素渲染。具体而言，NeRF 将空间位置 $\mathbf{x} \in \mathbb{R}^3$ 和视角方向 $\mathbf{d} \in \mathbb{R}^2$ 映射为密度 σ 和颜色 c ，*i.e.*, $f(\mathbf{x}, \mathbf{d}) = (\sigma, c)$ ，使用基于已知相机位姿图像训练的 MLP。为了渲染一个像素，NeRF 从相机向场景发射一条射线 $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ ，并在射线上以间隔 t_i 采样点。最终像素颜色 $C(\mathbf{r})$ 通过体素渲染计算得到：

$$C(\mathbf{r}) = \sum_i T_i \alpha_i \mathbf{c}_i, \quad \text{where } T_i = \exp\left(-\sum_{k=0}^{i-1} \sigma_k \delta_k\right), \tag{8}$$

这里的术语 $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$ 表示采样点的不透明度，累积透光率 T_i 量化了光线从 t_0 到 t_i 传播过程中不被任何粒子遮挡的概率。术语 $\delta_i = t_i - t_{i-1}$ 表示沿射线连续采样点之间的距离。

(3) 混合表示。当前大多数隐式方法依赖于回归 NeRF 或 SDF 值，这可能限制了其利用目标视图或表面显式监督的能力。然而，显式表示在训练过程中提供了有用的约束，并改善了用户交互。为了充分利用两种范式的互补优势，混合表示可被视为显式与隐式表示之间的一种权衡。

(i) 混合体素网格可用作方法中如 [124]–[126] 的混合表示。[125] 采用密度网格和特征网格进行辐射场重构，而 Instant-NGP [126] 使用基于哈希的多级网格，优化 GPU 性能以实现更快的训练和渲染。

(ii) DMTet [119] 将四面体网格与隐式 SDF 结合，实现灵活的三维表面表示。神经网络为每个顶点预测 SDF 值和位置偏移，从而能够建模复杂的拓扑结构。

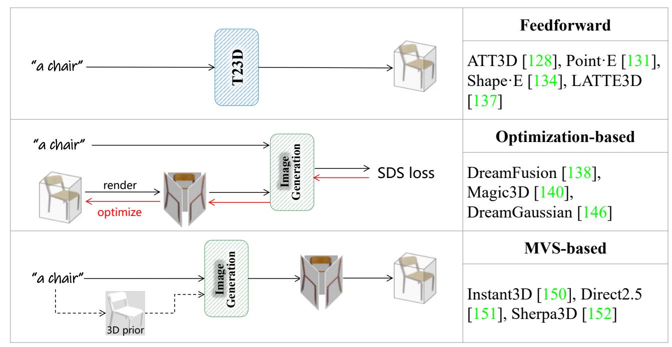

图 5: 不同文本生成 3D 范式之间的比较。图像生成模型是构建文本到 3D 方法的关键组成部分。

通过可微分的分割四面体（MT）层，将网格转换为网格模型，实现了高效且高分辨率的渲染。通过基于网格的损失优化几何和拓扑，DMTet 能够获得更精细的细节、更少的伪影，并在从粗粒度体素生成复杂三维数据集的条件形状合成任务中优于以往的方法。

(iii) 三平面为三维形状表示和神经渲染提供了一种内存高效的替代方案，用于体素网格。它将三维体积分解为三个正交的二维特征平面 (XY, XZ, YZ)。EG3D [127] 采用了这种结构，利用多层感知机（MLP）聚合平面中的特征，并在任意三维点上预测颜色和密度值：$(\sigma, c) = \mathtt{MLP}\big(\mathbf{f}_{xy}(\mathbf{x}) + \mathbf{f}_{xz}(\mathbf{x}) + \mathbf{f}_{yz}(\mathbf{x})\big)$。该方法相比基于体素的 NeRF 减少了内存消耗，并实现了更快的渲染速度。

2) 算法: **(1) 文本到 3D 生成**。通过模拟现实世界中的几何结构，从文本提示生成 3D 内容已受到广泛研究，可分为三个分支。读者可参考 [128]–[130] 以获取该领域的更全面综述。不同方法分支的对比在图 5 中展示。如我们所见，图像生成模型在文本到 3D 方法中扮

演关键角色，用于提供监督（即 SDS 损失）或合成多视角图像，以实现更精确的 3D 生成。

(i) 前向传播方法。受文本到图像生成的启发，一类主要方法将现有的成功生成式模型扩展至直接从文本提示中通过一次前向传播合成 3D 表示。其成功的关键在于将 3D 几何编码为紧凑表示，并使其与相应的文本提示对齐。

Michelangelo [131] 首先构建一个变分自编码器（VAE）模型，将 3D 形状编码为潜在嵌入。然后，该嵌入与从语言和图像中提取的嵌入进行对齐，使用 CLIP [30] 模型。通过对比损失进行最优化，可以从文本提示中推断出 3D 形状。ATT3D [132] 使用 Instant-NGP 模型作为 3D 表示，并通过映射网络将其与文本嵌入连接起来。接着，从 Instant-NGP 模型渲染多视角图像，并使用 SDS 损失优化整个网络。受 ATT3D 的启发，Atom [133] 学习从文本嵌入预测三平面表示，并采用两阶段最优化策略。Hyperfields [134] 训练一个动态超网络，以记录从不同场景中学成的 NeRF 参数。

最近，扩散模型的出色表现激发了研究人员将其扩展至 3D 生成。早期方法专注于从文本提示中学习合成显式 3D 表示。具体而言，Point· E [135] 首先使用 GLIDE [136] 合成多个视图，然后将这些视图作为条件，利用扩散模型生成点云。随后，MeshDiffusion [137] 使用扩散模型建立从文本到网格的映射。后续方法尝试将扩散模型应用于隐式 3D 表示。Shap· E [138] 首先将 3D 内容映射为辐射场的参数，然后训练一个扩散模型以生成这些参数，条件为文本嵌入。3D-LDM [139] 使用 SDF 表示 3D 内容的几何结构，并训练一个扩散模型实现文本条件生成。类似地，Diffusion-SDF [140] 构建了一个 SDF 自编码器，结合体素化扩散模型，从文本提示生成体素化带符号距离（SDF）。LATTE3D [141] 分别开发了纹理网络和几何网络，用于生成条件于文本嵌入的 NeRF 和 SDF，随后通过 SDS 损失优化一个 3D 感知的扩散模型。

讨论。与基于最优化的方法相比，前馈方法具有更高的效率，并且能够在无需测试时最优化的情况下生成 3D 内容。然而，这些方法严重依赖于数据量，通常在结构和纹理细节方面表现较差。

(ii) 基于最优化的方法。在文本到图像生成的基础上，另一类方法通过利用强大的文本到图像生成式模型提供丰富的监督信号，对 3D 表示进行优化。

DreamFusion [142] 首次引入得分蒸馏采样（SDS）损失，利用文本提示生成的图像来优化神经辐射场（NeRF）。MVDream [143] 对多视角扩散模型进行微调，以生成具有跨视角一致性的多视角图像，从而训练一个 NeRF 来捕捉三维内容。Magic3D [144] 采用带纹理的网格表示三维物体，并使用 SDS 损失对网格进行优化。Dream3D [145] 首先从文本提示生成一张图像，随后利用该图像生成三维形状以初始化神经辐射场。接着，通过 CLIP 引导对 NeRF 进行优化。Fantasia3D [146] 进一步结合 DMTet 和 SDS 损失，从文本提示生成三维物体。ProlificDreamer [147] 提出变分得分蒸馏（VSD），用于建模三维表示的分布，并生成具有丰富细节的高质量结果。为解决多面性 Janus 问题，PI3D [148] 首先对文生图扩散模型进行微调，以生成伪图像。然后，利用这些图像和 SDS 损失生成三维形状。VP3D [149] 首先使用文生图扩散模型从文本提示生成高质量图像，随后以生成的图像和文本提示为条件，通过 SDS 损失优化三维表示。

随着 3D 高斯的显著进展，其在文本到 3D 生成领域得到了广泛研究。DreamGaussian [150] 首次采用扩散模型获取 3D 高斯，并使用 SDS 损失进行最优化。随后，从 3D 高斯中提取网格，并对纹理进行精炼以获得更高质量的内容。为促进收敛，GSGEN [151] 和 GaussianDreamer [152] 首先利用 Point· E 从文本提示生成点云，以初始化高斯的位置。然后，这些高斯通过 SDS 损失优化，以精炼其几何和外观。Sculpt3D [153] 引入了 3D 先验，通过在数据库中检索参考 3D 对象实现，可无缝集成至现有流水线中。

讨论. 由于文本到图像模型中蕴含丰富的知识，基于最优化的方法能够生成更精细的细节。然而，这些方法需要昂贵的每提示最优化，并且耗时较长。

(iii) 基于多视图立体（MVS）的方法。与其直接从文本提示生成三维表示，不如更好地利用文本到图像模型，人们进行了大量尝试来合成多视角图像以实现三维生成。

Instant3D [154] 首先微调文本到图像的扩散模型以生成四视角图像。随后，这些图像被输入到一个 Transformer 中以预测三平面表示。Direct2.5 [155] 在 2.5D 渲染图像和自然图像上微调一个多视角法线扩散模型。给定一个文本提示，Direct2.5 首先生成法线图，并通过可微分光栅化进行优化。然后，将最优法线图作为条件来合成多视角图像。Sherpa3D [156] 首先使用一个 3D 扩散模型从文本提示生成粗略的 3D 先验。接着，

	Time (↓)	Quality (↑)	Alignment (↑)
Magic3D [144]	40min	38.7	35.3
Fantasia3D [146]	45min	29.2	23.5
DreamFusion [142]	30min	24.9	24.0
ProlificDreamer [147]	240min	51.1	47.8
MVDream [143]	30min	53.2	42.3
DreamGaussian [150]	7min	19.9	19.8
VP3D [149]	-	54.8	52.2
GaussianDreamer [157]	-	54.0	-

表 III: 不同文本到 3D 方法的定量比较。结果来自 [157]。

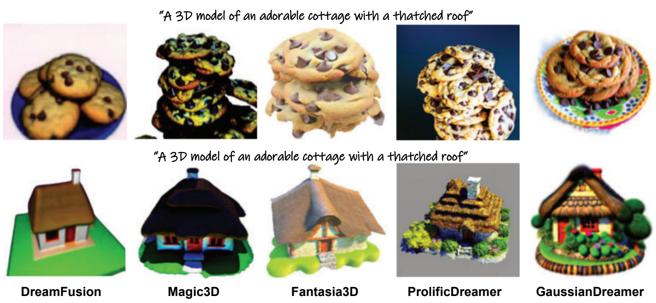

图 6: 不同文本到 3D 方法的定性比较。

生成法线图并用于合成具有 3D 一致性的多视角图像。

讨论. 随着视觉语言模型（VLMs）的最新进展，通过引入三维先验将这些二维生成模型扩展至三维生成已引起越来越多的关注。然而，三维一致性建模以及在有限三维数据下的微调问题仍待解决。

评估。文本到 3D 方法的定量评估仍然是一个开放性问题。对于主观质量评估，常用的基于参考的指标（如 PSNR）不适用，因为真实值数据不可用；而无参考质量指标（如 FID）可能并不总是与人类偏好一致。因此，大多数方法采用 CLIP 得分和 CLIP R-准确率来评估 3D 模型与文本提示的一致性。最近，多个基准 [158], [159] 已被建立，以全面评估文本到 3D 生成方法。在此，我们在表 III 中报告了代表性方法的定量得分，并在图 6 中展示了它们的视觉结果。读者可参考 [158], [159] 了解更多信息。

(2) 图像到 3D 生成。图像到 3D 任务的目标是生成与给定图像身份一致的高质量 3D 资产。由于 3D 数据采集成本较高，相较于图像和视频生成，文本到 3D 生成缺乏足够的高质量文本标注以实现规模化。由于图像天然包含了更丰富的低层信息，且与 3D 模态高度对齐，因此相比于文本到 3D 生成，图像到 3D 任务在输入与输出之间缩小了模态差距。因此，图像到 3D 生成已成为推动原生 3D 生成的基础性任务。为了利用图像生成模型内部的知识，它们通常被用作图像到 3D 模型的组成部分（图 8）。各方法的定性比较如图 9所示，定量比较如表 V所示。由于论文中使用的评估数据集或评价指标存在不一致性，部分工作未列于图和表中。

(i) 前馈方法。这些方法首先通过压缩网络（如变分自编码器，VAE）将 3D 资产编码为潜在向量，然后在潜在空间样本上训练生成式模型。3DGen [160] 引入三平面作为潜在空间，提升了压缩网络的准确率和效率。Direct3D [163] 采用三平面表示，并直接使用 3D 监督进行训练，保留了潜在三平面中的详细 3D 信息。Michelangelo [131] 受 3Dshape2vecset [184] 启发，利用 1D 向量作为潜在空间，并以占据场作为监督信号。CraftsMan [162] 进一步引入多视角生成模型，生成多视角图像作为扩散模型的条件，随后通过基于法线的优化对生成的网格进行精炼。Clay [161] 构建了一个在大规模 3D 数据集上预训练的综合性 3D 生成系统，包括基于 1D 向量的几何生成的变分自编码器与扩散模型、用于 PBR 材质的材质扩散模型，以及跨多种模态的条件设计。

讨论. 原生方法在 3D 数据集上训练压缩网络和生成模型，在几何生成方面表现出优于基于 MVS 和最优化的方法，能够生成更精细的几何细节。然而，由于制作和收集成本高昂，3D 数据集 [185], [186] 的增长速度远低于图像或视频数据集 [187], [188]。因此，原生方法缺乏足够多样且广泛的数据用于预训练。由此，如何利用来自视频和图像的先验信息来增强 3D 生成的多样性与泛化能力，特别是在纹理生成方面，仍是一个有待进一步探索的领域。

(ii) 基于最优化的方法。随着基于蒸馏的文本到 3D 模型方法的发展，基于最优化的方法通过由预训练图像到图像或文本到图像生成模型提供的 SDS 损失监督的训练过程，直接对 3D 资产进行最优化，并通过各种附加损失约束保持图像身份。

受 DreamFusion [142]、Magic3D [144] 和 SJC [189] 启发，RealFusion [165] 仅从预训练的文本到图像模型中提取先验，使用 SDS 损失，同时利用图像重构损失和文本反演分别保持低层特征和语义身份。随着大规模开放集 3D 数据集 [185] 的出现，Zero123 [166] 通过用新型视图合成模型替代文本到图像模型进行蒸馏，将 3D 数据集的先验引入图像到 3D 任务中。具体而言，Zero123 在 3D 数据集上微调一个预训练的图像到图像生成模型，引入相机位姿作为条件以控制生成图像的视角。该预训练的新型视图合成模型同时保留了图像模型

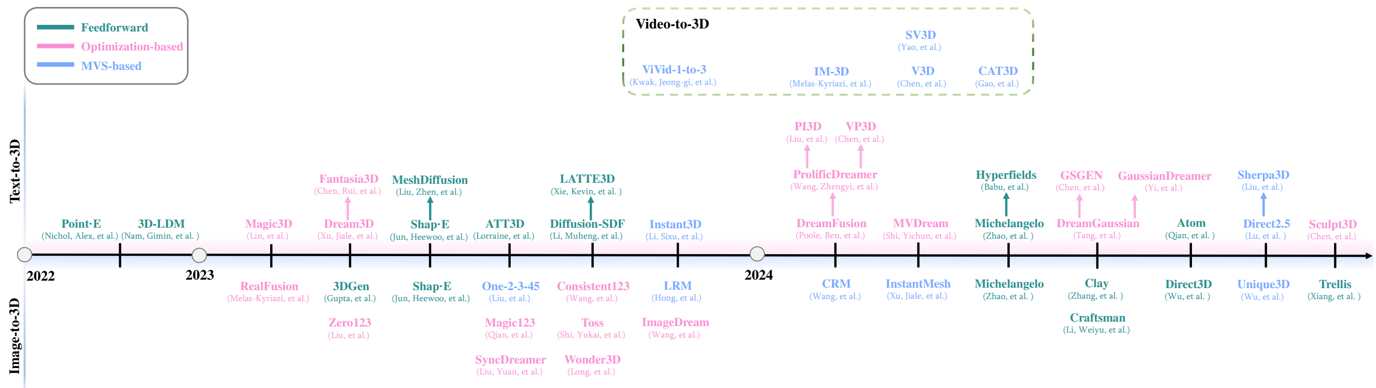

图 7: 近期文本到 3D、图像到 3D 以及视频到 3D 生成方法的时间线概述。

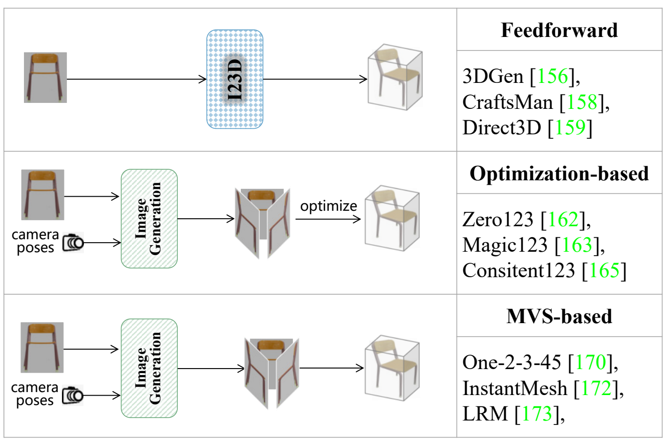

图 8: 不同图像到 3D 生成范式之间的比较。图像生成模型可用于合成多视角图像，以促进 3D 生成。

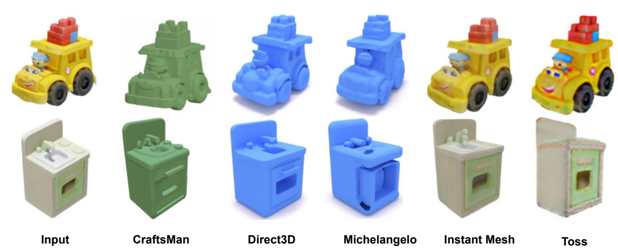

图 9: 不同图像到 3D 方法的定性比较。

中的细节以及 3D 数据集中的多视角一致性，显著缓解了 Janus 问题。

一系列工作扩展了 Zero123 [166]。Zero123-xl [186] 在更大的 3D 数据集（比原始数据集大 10× 倍）上预训练 Zero123 流水线，以提升泛化能力。Magic123 [167] 同时利用 2D 和 3D 先验进行蒸馏，以平衡泛化能力和一致性，并采用粗到细的流水线以获得更高品质。SyncDreamer [168] 和 Consistent123 [169] 均通过引入同步多视角扩散模型进一步提升 NVS 模型的多视角一致性：前者利用 3D 体素建模图像间的联合分布关系，后者则使用跨视角注意力和共享自注意力机制。Toss [170] 额外将文本描述作为 3D 数据的高层语义引入 NVS 模型预训练中，从而增强不可见视角的合理性与可控性。ImageDream [171] 通过设计多级图像提示控制器并结合文本描述进行训练，同时解决多视角一致性和 3D 细节问题。Wonder3D [173] 引入跨领域注意力机制，使 NVS 模型在优化过程中能同时去噪图像并对齐法向图；此外还额外引入法向图用于优化过程。IPDreamer [172] 通过引入 IPSDS（SDS 的一种变体）和掩码引导的对齐策略，实现从复杂图像提示中可控的 3D 合成，以保证多提示间的一致性。

讨论. 借助图像生成模型的强大先验，基于最优化的方法展现出强大的泛化能力，能够建模高精度的纹理。然而，由于新视角合成（NVS）模型在预训练阶段仅使用从 3D 数据中采样的 2D 数据进行监督，而非直接使用 3D 数据，多视角一致性问题无法从根本上解决，尽管通过 3D 体素建模或跨视角注意力有所改善。因此，基于最优化的方法通常面临几何过于平滑以及训练时间过长的问题，这源于其最优化范式。

(iii) 基于 MVS 的方法。基于 MVS 的方法将图像到 3D 生成过程分为两个阶段：首先使用一个 NVS 模型从单张图像生成多视角图像，然后直接利用前馈重构网络从这些多视角图像中创建 3D 资产。

基于 Zero123 [166] 预测的多视角图像，One-2-3-45 [174] 提出一个高程估计模块，并利用在 3D 数据集上预训练的基于 SDF 的通用神经表面重构模块进行 360° 网格重构，将重构时间缩短至 45 秒，相比基于最优化的方法。CRM [175] 进一步将多视角生成模型的输出图像固定在六个固定的相机位姿上，显著提升了多视角之间的一致性。随后，CRM 将多视角图像输入卷

积 U-Net，以深度图和 RGB 图像为监督信号，生成高分辨率三平面表示。InstantMesh [176] 同样将多视角图像的相机位姿固定，但采用基于 LRM [177] 的基于 Transformer 的多视角重构模型来重建 3D 网格，在牺牲部分图像到 3D 细节一致性的同时，提供了更好的泛化能力。Unique3d [178] 引入多级上采样策略，逐步生成更高分辨率的多视角图像，并采用法向图扩散模型预测多视角法向图，用于粗略网格的初始化，再基于多视角图像对网格进行精炼与着色。

讨论. 与基于最优化的方法相比，基于多视图立体（MVS）的方法在三维数据集上训练一个前馈重构模型，从多视角图像中重构出高质量的三维内容，显著提升了三维一致性，并将推理时间降低至秒级。然而，由于模型规模的限制，基于 MVS 的方法往往缺乏高质量的几何细节。

(3) 视频到 3D 生成。视频到 3D 的方法从根本上建立在 2D 扩散模型的进展之上，将基于图像的生成先验扩展至时序域。通过建模连贯的帧序列——具备一致的纹理、光照和几何结构——视频扩散模型能够利用 2D 中捕捉到的运动和视角变化实现隐式的 3D 结构学习。诸如 SV3D、Hi3D 和 V3D 等框架将预训练的视频扩散主干网络适配为生成多视角帧序列，随后通过体素渲染、网格优化或高斯 Splatting 进行整合，以重构显式的 3D 几何结构。这种协同作用充分利用了 2D 视频模型强大的内容合成能力，同时引入了空间一致性与相机控制，这对于稠密 3D 重构至关重要。在线视频数据的海量语料库构成了丰富的 3D 信息资源，捕捉了物体运动、视角变化以及相机转移，揭示了静态图像中通常无法获得的多视角视图 [190]–[195]。这些动态内容在连续帧之间提供了时间连贯性与空间一致性，对于理解复杂 3D 场景并生成高保真 3D 结构至关重要 [13]。因此，利用这些多视角且随时间变化的数据已成为重建和合成 3D 一致对象的一种有前景的方法 [196]。近期研究探索了基于视频的先验以实现鲁棒的 3D 生成 [180]–[182]，旨在学习在不同帧间保持一致且能适应视角变化的 3D 表示。从高层次来看，这些工作在视频到 3D 生成中的核心思想是使一个可控制相机的视频模型成为一致的多视角生成器，用于稠密 3D 重构（图 10）。

近年来，视频扩散模型的进展揭示了其在生成逼真视频方面卓越的能力，同时还能对三维结构进行隐式推理。然而，在利用这些模型进行有效三维生成方面仍存在显著挑战，尤其是在精确的相机控制方面。传统模型 [197]–[199] 通常局限于生成具有平滑且短距离相机轨迹的视频片段，这限制了它们创建动态三维场景或有效整合不同相机角度的能力。为解决这些局限性，已开发出多种创新技术以增强视频扩散框架中的相机控制能力。一种早期的方法是 AnimateDiff [58]，它采用低秩适应（LoRA） [200] 对视频扩散模型进行微调，以固定类型的相机运动。该方法能够在遵循指定相机动力学的同时，实现结构化场景的合成。另一项重要进展是 MotionCtrl [201]，它引入了条件机制，使模型能够遵循任意相机路径，从而在生成多样化视角方面实现更大的灵活性，并克服了先前方法的僵硬性。基于可控制相机的视频生成能力，SVD-MV [202]、SV3D [181] 和 IM-3D [179] 探索如何利用相机控制来提升从视频数据生成三维物体的效果。例如，SV3D 训练了一个能够渲染任意视角的视频扩散模型，展示了更强的泛化能力和高分辨率输出（576×576 像素）。这种能力使得在帧间保持空间一致性的同时适应各种视角成为可能，有效解决了密集重构中的关键挑战。尽管如此，这些方法通常将相机运动限制在围绕中心物体的固定轨道路径上，这限制了其在具有丰富上下文背景的复杂场景中的应用。然而，当需要生成复杂环境的引人注目的三维表示时，许多这些方法仍然表现不足，因为在这些环境中，变化的相机角度以及与多个物体的交互至关重要。

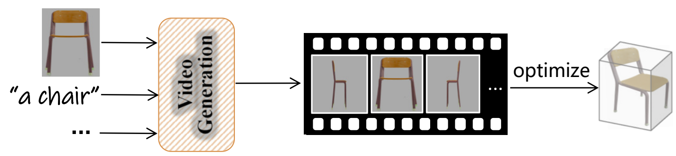

图 10: 视频生成 3D 的示意图。可控制相机的视频模型可作为 3D 生成的一致多视角生成器。

由于控制视频模型中的相机运动可以补充新视角信息，因此已有多种方法探索了视频扩散模型在新视角合成（NVS）中的潜力。例如，Vivid-1-to-3 [101] 有效融合了视图条件扩散模型与视频扩散模型，能够生成时间上一致的视角。通过确保帧间过渡平滑，该模型提升了合成输出的质量，特别适用于三维场景表示。CAT3D [182] 则利用多视图扩散模型增强了丰富的多视角信息。

讨论. 在多视角生成中利用视频先验，可将视频扩散模型转化为一致的多视角生成器，用于稠密 3D 重构。进一步探索将提升高保真度 3D 表示能力，尤其在需要强大多视角合成能力的复杂动态环境中。

3) 应用: **(1) Avatar Generation.** 随着元宇宙的兴起和 VR/AR 的普及，3D 头像生成引起了越来越多的关注。早期的工作集中在生成头部头像 [203]–[205]，这些方法利用文本到图像扩散模型和神经辐射场来创建面部资产。后续的方法更多关注通过将神经辐射场与统计模型结合来生成逼真的全身头像 [206], [207]。最近，头像生成的动画能力获得了极大的关注，许多方法被提出以实现这一目标 [208], [209]。

(2) Scene Generation. 除了头像生成外，场景生成以创建逼真的 3D 环境在元宇宙和具身智能等应用中也备受需求。早期方法聚焦于以物体为中心的场景，并利用条件扩散模型合成多视角图像来优化神经辐射场 [139], [140]。后续工作通过引入渐进策略将这些方法扩展到房间尺度的场景 [210], [211]。受其成功的启发，近期研究进一步探索了户外场景的生成，涵盖从街道尺度 [212], [213] 到城市尺度 [214], [215] 的范围。

(3) 3D Editing. 强大的 3D 生成能力已经催生了下游的 3D 内容编辑应用。一些方法专注于全局改变 3D 内容的外观或几何结构 [216], [217]，而没有将场景中的特定区域隔离出来。例如，场景风格化方法 [218], [219] 旨在操纵 3D 资产的风格，如光照调节和气候改变。最近的研究致力于实现更细粒度的 3D 内容编辑。具体来说，外观变化 [220], [221]、几何形变 [222], [223] 和对象级操作 [224], [225] 已经被研究，并取得了令人鼓舞的编辑效果。

D. 四维生成

我们通过整合所有维度，最终实现 4D 生成。作为计算机视觉领域的前沿方向，4D 生成专注于基于文本、图像或视频等多模态输入，合成随时间演变的动态 3D 场景。与传统的 2D 或 3D 生成不同 [226]，4D 合成引入了独特的挑战，要求在保持空间一致性的同时确保时间连贯性，并在高保真度、计算效率和动态真实感之间取得平衡 [227]。在本节中，我们首先介绍 4D 表示，其建立在 3D 表示的基础之上，随后总结当前的 4D 生成方法。近期研究探索了两种主要范式：基于最优化的方法利用得分蒸馏采样（Score Distillation Sampling, SDS），以及避免逐提示优化的前馈式方法。这两种范式针对不同的技术挑战，凸显了该领域的复杂性以及在视觉质量、计算效率与场景灵活性之间寻求可行平衡的持续努力。4D 生成领域的代表性工作汇总于表 VI。

1) 4D 表示: 4D 表示领域通过将时间维度引入 3D 建模，为理解动态场景提供了坚实基础。通过将静态的 3D 空间表示 (x, y, z) 扩展为包含时间 (t)，这些方法编码了场景动态和变换，这对于非刚性人体运动捕捉和模拟物体轨迹等应用至关重要 [257]–[260]。大多数 4D 表示可以分解为正则 3D 表示和形变两部分。第一模块正则 3D 表示用于建模静态模板形状；第二模块形变用于将模板形状变形以合成运动。为了对正则 3D 表示进行形变，常见的形变表示包括形变场和形变基元。形变场是一种神经网络，它将时空点映射到正则模板形状上的对应位置；形变基元如线性混合皮肤（linear blend skinning）则通过不同身体部位或控制点关联的刚性运动组合来表示某点的运动。这两种形变表示各有优劣。形变场比形变基元更灵活，这意味着形变场更具通用性，理论上可以拟合复杂的运动。而专为关节物体（如人体或动物）设计的形变基元在大范围关节运动中具有鲁棒性优势，因为形变场缺乏准确重构所需的归纳偏置，尤其是在快速运动时表现不佳。以下我们将重点聚焦于带有形变场类型的 4D 表示的 3D 正则表示。

4D 表示中的一个主要挑战是重建单个场景的计算成本过高。为了解决这一问题，显式方法和混合方法在不牺牲质量的前提下提升了效率。例如，平面分解通过将 4D 时空网格拆分为更小的组件来简化计算 [261]–[263]，而基于哈希的表示方法则降低了内存和处理需求 [264]。3DGS 通过使用变形网络将静态高斯分布动态调整，实现了速度与质量之间的平衡 [116], [265]。

最近的进展将静态与动态场景成分解，以高效地渲染刚性与非刚性运动。例如，D-NeRF 首先将场景编码到正则空间，然后将其映射到时序变形状态 [266]。3D Cinemagraphy 从单张图像生成基于特征的点云，并利用三维场景光流进行动画生成 [267]。4DGS 通过将尺度、位置和旋转等属性建模为时间函数来捕捉时序动态，同时保持静态场景不变 [268]。

基于混合 NeRF 的方法通过平面和体素特征网格扩展了 4D 建模。这些网格与 MLP 结合，实现了高效的全新视角合成，并通过引入时间平面扩展到动态场景 [261], [262]。可变形 NeRF 将几何结构与运动分离，简化了运动学习过程，并支持图像到 4D 视频生成、多视角重构等应用 [126]。总体而言，这些进展反映了在实现计算高效且高质量的动态场景时序建模方面持续的进步。

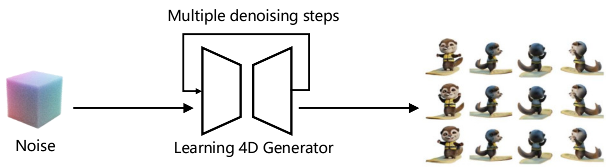

图 11: 4D 生成前向传播范式的示意图。我们直接在多视角动态数据上训练 4D 生成器。

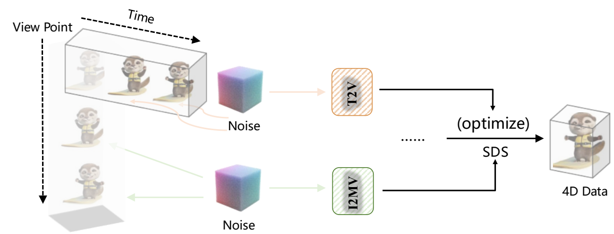

图 12: 基于 4D 生成最优化的范式示意图。我们使用 SDS 引导，使优化后的 4D 资产遵循预训练的文本到图像、多视角图像和文本到视频先验。

2) 算法: 现代的 4D 生成方法在很大程度上根植于 3D 生成所奠定的基础。特别是 3D 重构技术（如 NeRF 和 3DGS）的进步，直接推动了我们对动态 4D 场景的建模与渲染方式。这些 3D 框架不仅提供了高效的数据结构和渲染技术，还带来了对时间建模至关重要的归纳偏置。在表示层面，3D 方法提供了正则的空间先验，可通过形变场或运动轨迹扩展以捕捉时间演化。从训练角度来看，来自 3D 的快速训练技术（例如，哈希编码、层级采样）已被适配以加速 4D 最优化过程。人体动画作为最具代表性的 4D 任务，尤其受益于 3D 人体建模的进展与突破。诸如 SMPL(-X)、线性混合皮肤和神经形变场等技术，为关节运动建模提供了强大的结构先验。这些基于 3D 的工具显著提升了 4D 人体运动合成的真实感与可控性。

(1) Feedforward Approaches. 基于前馈的方法通过单次前向传播生成 4D 内容（图 11），避免了基于 SDS 的流水线所需的迭代最优化。这些方法依赖预训练模型，利用时间和空间先验实现快速且一致的生成。Control4D [228] 和 Animate3D [229] 直接从文本或视觉输入合成动态场景，支持交互媒体和个性化内容创作等实时应用。Vidu4D [230] 通过引入时间先验改进运动轨迹，确保帧间一致性与平滑转移。Diffusion4D [231] 通过结合时空特征抽取与高效的推理机制，将扩散模型的能力扩展到 4D 场景合成。L4GM [232] 进一步通过整合潜在几何建模增强了前馈技术，在保持计算效率的同时生成高质量结果。

讨论. 前馈式方法在强调速度和适应性的场景中表现优异，例如实时内容生成以及在消费级设备上的轻量级部署。然而，其对预训练模型的依赖以及在处理复杂动态时灵活性有限，使得在细节和多样性方面难以达到最优化方法的水平。尽管存在这些局限性，前馈技术仍是实现实际 4D 生成的重要进展，有效解决了计算效率和可扩展性等关键挑战。通过在质量和速度之间建立桥梁，这些方法有望在广泛的应用领域中推动 4D 内容生成的发展。

(2) Optimization-based Approaches. 基于最优化的方法在 4D 生成中具有基础性作用，采用迭代技术如得分蒸馏采样（SDS）将预训练扩散模型适配到动态 4D 场景的合成中（图 12）。这些方法利用文本到图像、多视角图像以及文本到视频生成模型的强大先验，实现具有丰富运动动态的时间一致性场景。例如，MAV3D [237] 通过文本提示引导的 SDS 损失对 NeRF 或 HexPlane 特征进行优化，而 4D-fy [238] 和 Dream-in-4D [241] 通过在 SDS 监督中整合图像、多视角和视频扩散模型来提升 3D 一致性和运动动态。AYG [239] 提出使用可变形 3DGS 作为内在表示，可以通过简单的 delta 形变场轻松解耦静态几何与动态运动，从而提高灵活性。

基于此类流水线，近期工作从多个方面进一步提升了 4D 生成的质量：外观保真度、几何一致性、运动忠实性以及生成可控性。特别地，TC4D [242] 和 SC4D [246] 实现了用户对 4D 物体运动轨迹的自由控制。STAG4D [248] 采用多视角融合策略，增强帧间空间与时间对齐效果，确保平滑的转移和一致性。此外，DreamScene4D [249] 和 DreamMesh4D [251] 采用解耦策略以定位优化重点，显著降低计算开销的同时保持高保真度。诸如 4Real [243] 和 C3V [244] 等最新进展通过将组合式场景生成与高效最优化相结合，进一步拓展了基于最优化方法的边界。这些方法将动态场景分解为模块化组件，如静态几何结构和运动场，从而实现灵活更新与多样内容生成。尽管这些方法在实现高质量且时间一致的结果方面表现出色，但基于最优化的方法仍存在计算开销大的问题，运行时需求通常难以支持实时应用。随着研究不断推进，当前工作持续聚焦于提升可扩展性并降低延迟，同时不牺牲视觉保真度或动态真实感。

评估。4D 生成方法的定量评估主要关注三个方面：1）生成单个 4D 资产所需的时间成本。2）CLIP 得分，

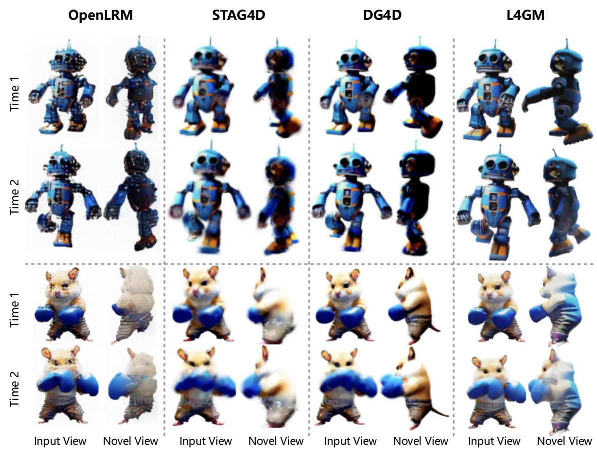

图 13: 不同 4D 生成方法的定性比较。结果来自 L4GM [232]。

用于评估生成的新视角与真实值之间的图像相似度。3) 生成结果与真实值新视角之间的感知相似度（LPIPS）。此处，我们在表VII中报告了代表性方法的定量得分，并在图 13中展示了它们的视觉结果。读者可参考 [232] 获取更多详细信息。

3) 应用: **(1) 4D Editing.** 指令引导的编辑，允许用户通过自然语言编辑场景，提供了一种用户友好且直观的方法。尽管这一方法已在 2D 图像上取得成功，例如使用 Instruct-Pix2Pix (IP2P) [271] 和 Instruct-NeRF2NeRF (IN2N) [217] 对 3D 场景进行编辑，但将其扩展到 4D 场景仍面临重大挑战。最近在文本到图像扩散模型和可微分场景表示方面的进展，使得利用文本提示编辑 4D 场景成为可能。例如，Instruct 4D-to-4D [272] 将 4D 场景视为伪 3D 场景，采用视频编辑方法迭代生成连贯的编辑数据集。同时，Control4D [228] 结合 GAN 与扩散模型，基于文本指令确保对动态 4D 肖像的一致编辑。

(2) Human Animation. 作为 4D 仿真的重要组成部分，人体动作生成是学术界关注度最高的研究方向之一。与第 III-B2 节中所述的人体中心视频生成不同，3D 人体动作生成在 3D 应用（如游戏和具身智能）中更易于实现角色动画。近期 3D 人体动作生成的成功主要得益于已深入研究的人体参数化模型 [273], [274]。

人体动作生成旨在实现一个基本目标：在数字世界中模拟 4D 人体主体，该任务可划分为两个方面。1) *通过稀疏控制信号生成动作主要基于用户指定的稀疏动作，在虚拟世界中模拟人体动画*。Robust motion in-between [275] 提出了一种时间到达嵌入和预定目标噪声向量，以在模型中鲁棒地完成不同转移长度的动作插值。由于动作空间的相位流形具有良好的结构，Starke *et al.* [276] 提出了专家网络混合模型，用于在相位流形上进行动作插值。此外，另一部分稀疏控制引导的动作生成是动作预测，*即动作外推法*。早期的动作预测研究 [277]–[280] 试图以确定性方式预测动作。考虑到动作预测具有主观性，一些工作 [281]–[285] 提出在预测中生成多样化的动作。2) *通过多模态条件生成动作旨在利用其他模态输入（如文本、音频、音乐）来模拟人体动作*。为解决成对文本-动作数据有限的问题，Guo *et al.* [286] 提出一个相对较大的文本-动作数据集 HumanML3D，其规模显著大于以往数据集，极大地推动了该任务的发展。此外，一些研究人员验证了 VQ-VAE [286]–[291] 是另一种有效的文本驱动动作合成方法。随着扩散模型的快速发展，大量工作 [292]–[297] 将扩散模型引入该任务，并取得了良好的生成质量。与文本到动作任务类似 [298]–[303]，音乐驱动舞蹈生成的技术路线也可分为 cVAE [304]、VQ-VQE [305] 和基于扩散模型 [306], [307] 的方法。

IV. 数据集和评估

在本节中，我们总结了用于二维、视频、三维和四维生成的常用数据集，如表 VIII 所示。随后，我们在表 IX 中呈现了一个统一且全面的评估指标总结。对于定量分析，我们从两个角度评估指标：1) *质量*：评估合成数据的感知质量，独立于输入条件（例如文本提示）。2) *对齐度*：衡量条件一致性，即生成数据与用户意图输入的匹配程度。对于定性分析，生成结果的视觉质量在评估方法中起着关键作用。因此，我们引入了一些基于人类偏好的指标作为参考，以更有效地开展用户研究，从而获得更具说服力的定性分析结果。此外，我们提倡强调部署生成式模型时面临的实际挑战，尤其是与计算效率相关的问题。许多最先进的方法需要大量 GPU 资源和较长的推理时间，这阻碍了它们在真实应用场景中的可访问性和可扩展性。这些因素往往未在评估指标中体现，但在考虑资源受限环境或交互式系统中的部署时至关重要。我们还建议未来的基准应包含运行时间、内存使用量和训练成本，以更好地反映生成式模型的实际可行性。

V. 未来方向

尽管 2D、视频、3D 和 4D 生成技术取得了快速进展，但许多开放性问题仍然存在，尤其是在多种模态

相互作用时。由于整合空间与时间维度的复杂性，这些难题更加突出。在 2D 生成中，提升生成图像的真实感和多样性仍是关键障碍。在视频生成中，建模长期时间动态以及确保帧间平滑过渡是至关重要的挑战。在 3D 生成中，如何在高质量输出与计算效率之间取得平衡仍然是核心问题。解决这些问题对于推进 4D 生成至关重要，因为 4D 生成建立在这些已有原则的基础之上。因此，制定清晰的发展路线图不仅有助于 4D 研究，也对 4D 模型所依赖的相关 2D、视频和 3D 领域具有重要意义。

Multimodal Generation: 生成多样且逼真的四维内容具有挑战性，因为真实场景在空间、时间以及感官通道上跨越了多种模式。多模态学习的研究揭示了三个持续存在的障碍。首先，需要跨模态的可靠对齐，以确保合成的四维资产能够遵循给定的文本、图像或视频提示。其次，高质量标注的多模态数据集稀缺，尤其是在特定领域，这限制了当前模型所能学习的真实世界动态的覆盖范围。第三，现有架构难以扩展至建模随时间演变的高分辨率三维场景所面临的内存和计算代价。条件扩散、组合潜在空间以及跨模态对比预训练虽提升了生成质量，但在这些约束下同时实现多样性和保真度仍是开放的研究问题。

Temporal Consistency and Coherence: 在 4D 生成中，确保帧与帧之间随时间的平滑且真实的转移是一个重大挑战。与静态的 3D 生成不同，4D 生成需要在时间步上保持形状、纹理和运动的一致性。在长序列中，闪烁或不自然的形变等伪影很容易出现。如何在不牺牲细节或真实感的前提下，实现时间一致性，仍然是一个未解决的问题。

Efficiency and Controllability: 生成 4D 资产涉及大规模时空张量和漫长的最优化周期，对 GPU 内存和计算资源造成巨大压力。当使用得分蒸馏采样时，这些成本进一步增加，因为其隐式公式使结果对控制信号和扩散先验高度敏感。需要紧凑的表示方法和可扩展的架构来降低这些负担。同时，当前流水线仍仅支持粗粒度控制；精确的运动路径和语义编辑仍然困难。进展需要面向控制的算法和清晰的交互界面。

Fidelity and Diversity: 在 4D 生成中同时保持保真度和多样性仍然具有挑战性。对于保真度而言，形状、纹理和运动必须在时间上保持一致；否则，长序列会受到闪烁、细节模糊以及几何或光度漂移的影响。在图像到 4D、视频到 4D 以及 3D 到 4D 生成流水线中使用的扩散模型难以从有限输入中推断缺失视角和高频细节，这凸显了对更强的时间正则化、视图感知条件控制以及内存高效的架构的需求，以在确保平滑演化的同时保持每帧的质量。对于多样性而言，当前模型在不同物体、运动风格和环境下的泛化能力较差，因为动态 3D 内容变化极大，而训练数据集的覆盖范围仍然有限。为了实现稳健的泛化，需要更广泛的训练数据覆盖、自适应条件控制以及模块化训练策略，从而即使在未见过的场景中也能保持高质量的合成效果。

Physics and Dynamics Modeling: 真实感 4D 生成必须以物理准确性再现碰撞、弹性和塑性变形以及流体流动，但将此类物理嵌入生成式模型具有挑战性，因为这涉及求解偏微分方程或实时模拟交互。这使得在物理保真度和计算速度之间找到可行的平衡变得困难。一个有前景的方向是将生成网络与可微分模拟器或神经普通微分方程层相结合，使基于梯度的训练能够强制执行牛顿定律。构建这样的耦合系统构成了一个封装现实世界特征的世界模型的基础，为评估物理和动态一致的 4D 生成提供了一个自然的基准。未来的研究可以在此类基准之上，更好地模拟真实交互，同时保持计算效率。

VI. 结论

在本综述中，我们回顾了近年来用于模拟真实世界的多模态生成式模型的最新进展，重点关注外观、动态和几何之间的相互交织维度。我们对 2D、视频、3D 和 4D 生成领域的现有方法进行了分类，讨论了其代表性方法、跨领域关系及技术差异，并辅以对比性的视觉示例。此外，我们总结了常用的数据集和评估指标，为基准测试提供了实用参考。

尽管取得了快速进展，一些根本性挑战依然存在，例如生成流水线的可扩展性、长序列中的时间一致性以及对现实世界动态的适应能力。我们识别出若干开放性的研究方向，包括跨模态统一表示的需求、基于稀疏监督的高效训练，以及融入物理约束以提升真实感。

我们希望本综述不仅能为新手提供全面的概览，也能为未来研究更连贯、可控且具有物理基础的多模态生成系统奠定基础。

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表 IV: 最近的文本到 3D、图像到 3D 以及视频到 3D 生成方法。

Paradigms		Methods	Paper	Code	Representations	Objectives
Text-to-3D	Feedforward	Point·E [135]	Link	Link	Point Clouds	MAE
		3D-LDM [139]	Link	-	SDF	reconstruction+regularization
		ATT3D [132]	Link	Link	Instant-NGP model	SDS
		Diffusion-SDF [140]	Link	Link	SDF	reconstruction+KL-Divergence
		LATTE3D [141]	Link	Link	NeRF/SDF	SDS+regularization
		MeshDiffusion [137]	Link	Link	Mesh	diffusion loss
		Shap·E [138]	Link	Link	NeRF	rendering loss
		Hyperfields [134]	Link	Link	NeRF	SDS
		Michelangelo [131]	Link	Link	Occupancy Field	text-image-shape contrastive + BCE
		Atom [133]	Link	-	Triplane	SDS
	Optimization	Magic3D [144]	Link	Link	NeRF/DMTet	SDS
		Dream3D [145]	Link	-	NeRF	reconstruction+regularization
		Fantasia3D [146]	Link	Link	DMTet	SDS
		DreamFusion [142]	Link	-	NeRF	SDS
		ProlificDreamer [147]	Link	Link	NeRF	VSD
		PI3D [148]	Link	-	Triplane	SDS
		VP3D [149]	Link	-	NeRF	SDS
		MVDream [143]	Link	Link	NeRF	SDS
		DreamGuassian [150]	Link	Link	3DGS	SDS
		GaussianDreamer [157]	Link	Link	3DGS	SDS
		GSGEN [151]	Link	Link	3DGS	SDS
		Sculpt3D [153]	Link	Link	NeRF	diffusion loss
	MVS	Instant3D [154]	Link	Link	Triplane	MSE+LPIPS
		Direct2.5 [155]	Link	Link	Multi-view normal maps	normal rendering+alpha mask loss
		Sherpa3D [156]	Link	Link	DMTet	SDS
Image-to-3D	Feedforward	3DGen [160]	Link	-	DMTet	rendering loss
		Shap·E [138]	Link	Link	NeRF	rendering loss
		Michelangelo [131]	Link	Link	Occupancy Field	text-image-shape contrastive + BCE
		Clay [161]	Link	Link	Occupancy Field	BCE
		CraftsMan [162]	Link	Link	Occupancy Field	BCE
		Direct3D [163]	Link	Link	Occupancy Field	BCE
		Trellis [164]	Link	Link	3DGS/RF/Mesh	MSE
	Optimization	RealFusion [165]	Link	Link	NeRF	SDS/Image Reconstruction
		Zero123 [166]	Link	Link	NeRF	SDS
		Magic123 [167]	Link	Link	NeRF	SDS
		Syncdreamer [168]	Link	Link	NeRF	SDS
		Consistent123 [169]	Link	Link	NeRF	SDS
		Toss [170]	Link	Link	NeRF	SDS
		ImageDream [171]	Link	Link	NeRF	SDS
		IPDreamer [172]	Link	Link	NeRF	SDS
		Wonder3D [173]	Link	Link	Mesh	Image/Normal MSE
	MVS	One-2-3-45 [174]	Link	Link	SDF	Image/Depth MSE
		CRM [175]	Link	Link	Triplane	Image/Depth MSE
		InstantMesh [176]	Link	Link	Triplane	Image/Depth/Normal MSE
		LRM [177]	Link	Link	NeRF	MSE
		Unique3D [178]	Link	Link	Mesh	Image/Normal MSE
Video-to-3D	MVS	ViVid-1-to-3 [101]	Link	Link	Multi-view images	-
		IM-3D [179]	Link	Link	3DGS	MSE+LPIPS
		V3D [180]	Link	Link	Mesh	MSE+LPIPS
		SV3D [181]	Link	Link	NeRF / DMTet	MSE+LPIPS
		CAT3D [182]	Link	Link	NeRF	MSE+LPIPS

表 V: 与基准方法在 GSO 数据集上的定量比较 [183]。

Type	Method	CD↓	IoU↑	Time↓
Optimization-based Approaches	Realfusion [165]	0.0819	0.2741	~90min
	Magic123 [167]	0.0516	0.4528	~60min
MVS-based Approaches	One-2-3-45 [174]	0.0629	0.4086	~45s
	Zero123 [166]	0.0339	0.5035	~10min
	InstantMesh [176]	0.0187	0.6353	~10s
Feedforward Approaches	Point-E [135]	0.0426	0.2875	~40s
	Shap-E [138]	0.0436	0.3584	~10s
	Michelangelo [131]	0.0404	0.4002	~3s
	CraftsMan3D [162]	0.0291	0.5347	~5s

表 VI: 4D 生成方法的代表性作品。"Rep" 代表表示。

Approaches	Methods	Paper	Code	Representations	Priors / Models	Objectives
Feedforward	Control4D [228]	Link	-	GaussianPlanes	GAN-based	GAN objective
	Animate3D [229]	Link	Link	4DGS	Diffusion-based	latent diffusion loss
	Vidu4D [230]	Link	Link	Dynamic Gaussian Surfels	Diffusion-based	reconstruction/regularization loss
	Diffusion4D [231]	Link	Link	4DGS	Diffusion-based	latent diffusion + motion reconstruction loss
	L4GM [232]	Link	Link	3DGS	VAE-based	MSE+LPIPS
	GenXD [233]	Link	Link	4DGS	Diffusion-based	SSIM+LPIPS
	CAT4D [234]	Link	-	Deformable 3DGS	Diffusion-based	LPIPS
	4Real-Video [235]	Link	-	Deformable 3DGS	Diffusion-based	velocity matching loss of rectified flow [236]
Optimization	MAV3D [237]	Link	-	HexPlane-/NeRF-based 4D Rep	T2I/T2V	SDS
	4D-fy [238]	Link	Link	Hash-/NeRF-based 4D Rep	MVDream/T2I/T2V	Hybrid SDS
	AYG [239]	Link	-	Deformable 3DGS	MVDream, T2I/T2V	SDS variant(CSD [240])
	Dream-in-4D [241]	Link	Link	NeRF-based 4D Rep	MVDream/T2I/T2V	Hybrid SDS
	TC4D [242]	Link	Link	Hash-/NeRF-based 4D Rep	T2V	SDS
	4Real [243]	Link	-	Deformable 3DGS	T2V	SDS
	C3V [244]	Link	-	3DGS	LDMs [38]	SDS
	Consistent4D [245]	Link	Link	D-NeRF	Zero123	SDS
	SC4D [246]	Link	Link	SC-GS [247]	SVD/Zero123	SDS
	STAG4D [248]	Link	Link	Deformable 3DGS	SVD/Zero123	SDS
	DreamScene4D [249]	Link	Link	Deformable 3DGS	SVD/Zero123	SDS
	4DM [250]	Link	Link	D-NeRF	SVD	4D-aware SDS + anchor loss
	DreamMesh4D [251]	Link	Link	SC-GS+Mesh Rep	Zero123	SDS
	4K4DGen [252]	Link	Link	4DGS	SVD	L1
	EG4D [253]	Link	Link	4DGS	SVD	L1
	AvatarGO [254]	Link	Link	3DGS+Hexplane	SVD/Zero123	SDS
	Disco4D [255]	Link	Link	4DGS	SVD	SDS
	GenMOJO [256]	Link	Link	Deformable 3DGS	SVD/Zero123	SDS

	Time (↓)	CLIP (↑)	LPIPS (↓)
Consistent4D [245]	2hr	0.87	0.16
4DGen [269]	1hr	0.89	0.13
STAG4D [248]	1hr	0.91	0.13
DG4D [270]	10min	0.87	0.16

表 VII: 不同 4D 生成方法的定量比较。结果来自 [232]。

表 VIII: 常用二维、视频、三维和四维生成数据集的总结。[Link] 指向数据集网站。

Dataset	Year	# Categories	# Objects	# Images	Type	Evaluation Metrics	Contribution Highlights	Limitations
2D Generation (Sec. III-A)								
SBU [308] [Link]	2011	89	-	1M	Image-Text pairs	-	Photographs with captions and 5 kinds of image content.	Noisy user captions and single reference.
MS-COCO [309] [Link]	2014	80 for objects, 91 for stuffs	1.5M	330K	Image-Text pairs	FID, CLIP-Score, SSIM	Manual annotation and 330K for English.	Class imbalance and social bias.
CC-3M [310] [Link]	2018	-	-	3M	Image-Text pairs	-	Alt-text caption annotation.	Noisy user captions and weakly aligned.
LAION-5B [187] [Link]	2022	-	-	5.85B	Image-Text pairs	FID, CLIP-Score, SSIM	5.85B CLIP-filtered multilingual image-text pairs.	Uncurated and unsafe contents.
ShareGPT4V [311] [Link]	2023	-	-	102K	Image-Text pairs	-	1.2M image-text pairs for ShareGPT4V-PT.	AI generated captions with hallucination.
Pick-a-Pic [312] [Link]	2023	-	-	500K	Image-Text pairs	FID, PickScore	Human preferences on model-generated images.	Limited preference labels.
Video Generation (Sec. III-B)								
UCF-101 [313] [Link]	2012	101	-	13K	action recognition	FVD	Manual caption annotation.	Strong background bias misleads action recognition.
ActivityNet [314] [Link]	2015	200	-	28K	action recognition	FVD	Manual caption annotation.	Background bias.
MSR-VTT [315] [Link]	2016	20	-	10K	video-language dataset	FVD, CLIP-Score	Manual caption annotation. Resolution: 240p.	Noisy and duplicate captions.
HowTo100M [316] [Link]	2019	12	154	136M	video-language dataset	FVD, CLIP-Score	ASR annotation. Resolution: 240p.	Noisy and misaligned narration.
WebVideo-10M [317]	2021	-	-	10M	video-language dataset	FVD, CLIP-Score	Alt-text caption annotation. Resolution: 360p.	Low-resolution and watermarked videos.
HD-VILA-100M [318] [Link]	2022	15	-	103M	video-language dataset	FVD, CLIP-Score	ASR annotation. Resolution: 720p.	Noisy and misaligned ASR subtitles.
InternVid [319] [Link]	2023	16	-	7M	video-language dataset	FVD, CLIP-Score	Automatic caption annotation.	Limited diversity.
Panda-70M [188] [Link]	2024	-	-	70M	video-language dataset	FVD, CLIP-Score	Automatic caption annotation. Resolution: 720p.	Limited diversity.
Koala-36M [320] [Link]	2024	-	-	36M	video-language dataset	FVD, CLIP-Score	Automatic and manual caption annotation. Resolution: 720p.	Inherent biases from its source corpus.
3D Generation (Sec. III-C)								
DeepFashion [321] [Link]	2016	50	300K	800K	single-view images	-	Human clothing dataset.	No 3D data.
SHHQ [322] [Link]	2022	-	-	230K	single-view images	-	Human body dataset.	(1) No 3D data. (2) Inherent biases from FFHQ.
CO3D [323] [Link]	2021	50	19K	1.5M	multi-view images	PSNR/LPIPS; IoU, etc.	Annotated with camera poses and point clouds.	Lack high-fidelity geometry/textures.
RTMV [324] [Link]	2021	-	2K	300K	multi-view images	PSNR/LPIPS; MAE, etc.	Large-scale synthetic dataset with high-resolution images.	Synthetic data cannot fully capture real-world complexity.
MVImgNet [325] [Link]	2023	238	219K	6.5M	multi-view images	PSNR/LPIPS	Annotated with camera poses and point clouds.	Lack high-fidelity geometry/textures.
uCO3D [326] [Link]	2025	1K	170K	170K	multi-view images	PSNR/LPIPS; IoU	Annotated with camera poses, point clouds, depth maps, caption, and 3DGS.	Lack high-fidelity geometry/textures.
ShapeNet [327] [Link]	2015	3K	51K		3D data	-	Large-scale synthetic CAD dataset.	Synthetic data cannot fully capture real-world complexity.
3D-Future [328] [Link]	2020	34	10K	-	3D data	CLIP Similarity	High-quality 3D models with textures and attributes.	(1) Lack object diversity. (2) Synthetic data cannot fully capture real-world complexity.
GSO [183] [Link]	2022	17	1K	-	3D data	F-Score/ CLIP Similarity	High-fidelity geometry/textures	Limited to household objects.
Objaverse [185] [Link]	2022	21K	818K	-	3D data	F-Score/ CLIP Similarity	Large-scale dataset.	Variable data quality.
Objaverse-XL [186] [Link]	2023	-	10.2M	-	3D data	F-Score/ CLIP Similarity	Large scale and high diversity.	Variable data quality.
Cap3D [329] [Link]	2023	-	785k	-	3D-text pairs	FID/CLIP etc.	Rich text annotations	Inherent bias in captions from pretrained models.
Animal2400 [132]	2023	-	2400	-	prompts	CLIP R-Precision	10 animals, 8 activities, 6 themes, 5 hats.	Limited object diversity.
DF27/DF415 [142] [Link]	2024	-	27/415	-	prompts	CLIP R-Precision	-	Limited to synthetic or artist-created objects.
4D Generation (Sec. III-D)								
Consistent4D [245] [Link]	2023	-	-	12(in-the-wild), 14(synthetic)	multi-view videos	LPIPS/CLIP	*Video-to-4D Generation.*	-
Diffusion4D [231] [Link]	2024	-	365K	365K	dynamic 3D data	-	Collected from Objaverse-1.0 (42K) and Objaverse-xl (323K)	-
MV-Video [229] [Link]	2024	-	53K	53K	dynamic 3D data	-	Annotated by MiniGPT4-Video [330]	Inaccurate captions.
CamVid-30K [233] [Link]	2024	-	-	30K	4D data	-	Real-world 4D scene dataset	-
4D-DRESS [331] [Link]	2024	-	-	64(outfits)	human clothing 4D data	CD	Total number of 3D human frames: 78k.	Manual-heavy pipeline limits scalability.

”ASR” stands for automatic speech recognition, ”pcl” for point clouds, and ”CD” for Chamfer Distance.

表 IX: 常见评估指标概述。

	Notation	Description
		Quality (quantitative analysis)
Image-level	PSNR ↑	*Image Fidelity/Similarity.* **P**eak **S**ignal-to-**N**oise **R**atio: the ratio between the peak signal and the Mean Squared Error (MSE).
	SSIM ↑	*Image Fidelity/Similarity.* **S**tructural **S**imilarity **I**ndex **M**easure [332]: evaluating brightness, contrast, and structural features between generated and original images.
	LPIPS ↓	*Image Fidelity/Similarity.* **L**earned **P**erceptual **I**mage **P**atch **S**imilarity [333]: metric computed with a model trained on labeled human-judged perceptual similarity.
	FID ↓	*Image Fidelity/Similarity.* **F**réchet **I**nception **D**istance [84]: comparison of generated and GT image distributions.
	IS ↑	*Image Fidelity/Similarity.* **I**nception **S**core [334]: metric only evaluated image distributions and calculated by pretrained Inception v3 model.
Video-level	FVD ↓	*Video Fidelity.* **F**réchet **V**ideo **D**istance [335]: the Inflated-3D Convnets (I3D) pretrained model to compute their means and covariance matrices for scores.
	KVD ↓	*Video Fidelity.* **K**ernel **V**ideo **D**istance [335]: an alternative to FVD proposed in the same work, using a polynomial kernel.
	Video IS ↑	*Video Fidelity.* **Video I**nception **S**core [336]: the inception score of videos with the features extracted from C3D [337].
	FVMD ↓	*Video Fidelity.* **F**réchet **V**ideo **M**otion **D**istance [338]: metric focused on *temporal consistency*, measuring the similarity between motion features of generated and reference videos using Fréchet Distance.
		Alignment (quantitative analysis)
Semantics	CLIP Similarity [339] ↑	*Text-Image Alignment.* Same as CLIP-Score, defined as a text-to-image similarity metric, measuring the (cosine) similarity of the embedded image and text prompt.
	CLIP R-Precision [340] ↑	*Text-Image Alignment.* The CLIP model’ s accuracy at classifying the correct text input of a rendered image from amongst a set of distractor query prompts.
	X-CLIP [341] ↑	*Text-Image Alignment.* Measured by a video-based CLIP model finetuned on text-video data.
	CLIP-T ↑	*CLIP Textual Alignment.* The average cosine similarity between the generated frames and text prompt with CLIP ViT-B/32 [342] image and text models.
	CLIP-I ↑	*CLIP Image Alignment.* The average cosine similarity between the generated frames and subject images with CLIP ViT-B/32 image model.
	DINO-I ↑	*DINO [86] Image Alignment.* The average visual similarity between generated frames and reference images with DINO ViT-S/16 model.
		User study (qualitative analysis)
	AQ ↑	*Human Preference.* **A**ppearance **Q**uality: percentage unit (%).
	GQ ↑	*Human Preference.* **G**eometry **Q**uality: percentage unit (%).
	DQ ↑	*Human Preference.* **D**ynamics **Q**uality: percentage unit (%).
	TA ↑	*Human Preference.* **T**ext **A**lignment: percentage unit (%).